Monotonic Kolmogorov–Arnold Network (KAN) Transport Maps for L-SING: Theoretical Foundations and Monotonicity Proofs

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Abstract

In the Localized Sparsity Identification for Non-Gaussian Distributions (L-SING) framework [6], undirected graphical models are discovered by estimating local conditional transport maps $S^k(x_{-k}, \cdot)$ that push forward continuous conditional distributions $\pi(x_k \mid x_{-k})$ to a standard normal reference measure $\eta_k = \mathcal{N}(0, 1)$. Existing implementations parameterize $S^k(x)$ using Unconstrained Monotonic Neural Networks (UMNNs) [14], which guarantee target monotonicity $\partial_k S^k > 0$ via numerical quadrature (e.g., 50-node Clenshaw–Curtis grids), imposing heavy computational overhead and requiring higher-order automatic differentiation through the numerical integrator. In this work, we rigorously examine the replacement of UMNNs with Monotonic Kolmogorov–Arnold Networks (KANs). We systematically identify the monotonicity requirements dictated by measure transport, proving that strict target monotonicity is fully enforceable across single-layer, two-layer, and arbitrarily deep (L-layer) KANs, while retaining unconstrained expressivity along conditioning dimensions x_{-k}. Furthermore, we prove that Scale-Modulated Monotonic KANs (SM-KANs) strictly preserve target monotonicity while overcoming the inherent additive polarization dilemma of standard KANs on multiplicative non-Gaussian dependencies. We present both an intuitive, accessible explanation and formal mathematical proofs, connecting our results with the recently established MonoKAN framework [11] and Fritsch–Carlson monotonicity conditions [4].

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1 The Measure Transport Formulation in L-SING

Remark 1.1 (Context and Scope). *Note for collaborator:* For completeness and clarity of notation, the following subsections provide a self-contained re-derivation of the L-SING conditional pullback density formula and the resulting learning loss function.

1.1 Conditional Transport Maps

Let $\eta_k = \mathcal{N}(0,1)$ denote the standard univariate Gaussian reference measure with Lebesgue density,

$$\eta_k(z_k) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2} z_k^2\right). \tag{1}$$

For each node $k \in \{1, \dots, d\}$, L-SING seeks a conditional transport map component $S^k(x_{-k}, \cdot) : \mathbb{R} \to \mathbb{R}$, $x_k \mapsto z_k = S^k(x_{-k}, x_k)$, that pushes forward the conditional probability measure $\nu_{\pi(\cdot|x_{-k})}$ to the standard normal reference measure ν_{η_k},

$$S^k(x_{-k}, \cdot)_\sharp \, \nu_{\pi(\cdot|x_{-k})} = \nu_{\eta_k}. \tag{2}$$

By definition of the pushforward measure in set-integral form, for any Borel set $B \in \mathcal{B}(\mathbb{R})$,

$$\mathbb{P}\Big(Z_k \in B \;\Big|\; X_{-k} = x_{-k}\Big) = \mathbb{P}\Big(X_k \in (S^k(x_{-k}, \cdot))^{-1}(B) \;\Big|\; X_{-k} = x_{-k}\Big). \tag{3}$$

Expressed as Lebesgue integrals against their respective probability densities,

$$\int_B \eta_k(z_k)\, dz_k = \int_{(S^k(x_{-k},\cdot))^{-1}(B)} \pi(x_k \mid x_{-k})\, dx_k \quad \forall B \in \mathcal{B}(\mathbb{R}). \tag{4}$$

[Key Requirement: 1. Target Monotonicity $\partial_k S^k > 0$]
For the fiber map $x_k \mapsto S^k(x_{-k}, x_k)$ to be an orientation-preserving diffeomorphism on $\mathbb{R}$, its derivative along the target coordinate must be strictly positive everywhere,

$$\partial_k S^k(x) := \frac{\partial S^k(x)}{\partial x_k} > 0 \quad \forall x \in \mathbb{R}^d. \tag{5}$$

This ensures that the coordinate transformation $x_k \mapsto z_k$ is one-to-one, globally invertible on $\mathbb{R}$, and has a strictly positive Jacobian $|\partial_k S^k(x)| = \partial_k S^k(x) > 0$.

Under this strict monotonicity condition, we apply the 1D change-of-variables substitution $z_k = S^k(x_{-k}, x_k)$ to the right-hand side of (4). As x_k traverses $(S^k(x_{-k}, \cdot))^{-1}(B)$, z_k traverses B, with infinitesimal element $dz_k = \partial_k S^k(x_{-k}, x_k)\, dx_k$. Substituting $dx_k = \frac{1}{\partial_k S^k(x_{-k}, (S^k(x_{-k},\cdot))^{-1}(z_k))}\, dz_k$ yields,

$$\int_B \eta_k(z_k)\, dz_k = \int_B \frac{\pi\Big((S^k(x_{-k}, \cdot))^{-1}(z_k) \;\Big|\; x_{-k}\Big)}{\partial_k S^k\Big(x_{-k}, (S^k(x_{-k}, \cdot))^{-1}(z_k)\Big)}\, dz_k. \tag{6}$$

Rearranging into a single integral over B,

$$\int_B \left(\eta_k(z_k) - \frac{\pi\Big((S^k(x_{-k}, \cdot))^{-1}(z_k) \;\Big|\; x_{-k}\Big)}{\partial_k S^k\Big(x_{-k}, (S^k(x_{-k}, \cdot))^{-1}(z_k)\Big)}\right) dz_k = 0. \tag{7}$$

Since this identity must hold for **every** Borel set $B \in \mathcal{B}(\mathbb{R})$, the integrand vanishes Lebesgue-almost everywhere,

$$\eta_k(z_k) = \frac{\pi\Big((S^k(x_{-k},\cdot))^{-1}(z_k) \;\Big|\; x_{-k}\Big)}{\partial_k S^k\Big(x_{-k}, (S^k(x_{-k},\cdot))^{-1}(z_k)\Big)}. \tag{8}$$

Substituting $z_k = S^k(x)$ and solving for the conditional density yields the exact **conditional pullback density formula**,

$$\pi_{S^k}(x_k \mid x_{-k}) = \eta_k(S^k(x)) \cdot |\partial_k S^k(x)| = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}(S^k(x))^2\right) \cdot \partial_k S^k(x). \tag{9}$$

Taking the natural logarithm yields the conditional log-density of the model,

$$\log \pi_{S^k}(x_k \mid x_{-k}) = -\frac{1}{2}(S^k(x))^2 + \log \partial_k S^k(x) - \frac{1}{2}\log(2\pi). \tag{10}$$

1.2 Connection to the Learning Objective and Functional Regularization

The training objective in L-SING is derived directly from minimizing the Kullback–Leibler (KL) divergence between the true data distribution π and the model's pullback distribution π_{S^k},

$$D_{\mathrm{KL}}(\pi \,\|\, \pi_{S^k}) = \mathbb{E}_{X\sim\pi}\left[\log \pi(X_k \mid X_{-k}) - \log \pi_{S^k}(X_k \mid X_{-k})\right] \tag{11}$$

$$= -\mathbb{E}_{X\sim\pi}\left[\log \pi_{S^k}(X_k \mid X_{-k})\right] + \text{const}, \tag{12}$$

where the constant depends only on the true data entropy and does not involve the map parameters. Substituting the conditional log-pullback formula (10) into this expected negative log-likelihood directly yields,

$$-\mathbb{E}_{X\sim\pi}\left[\log \pi_{S^k}(X_k \mid X_{-k})\right] = \mathbb{E}_{X\sim\pi}\left[\frac{1}{2}(S^k(X))^2 - \log \partial_k S^k(X)\right] + \frac{1}{2}\log(2\pi). \tag{13}$$

Dropping the irrelevant additive constant $\frac{1}{2}\log(2\pi)$ and replacing the expectation with an empirical Monte Carlo average over M i.i.d. samples $\{x^{(i)}\}_{i=1}^M \sim \pi$, we obtain the empirical negative log-likelihood. To promote sparsity, a regularization term $\Phi(S^k)$ is added, leading to the regularized loss function,

$$\mathcal{L}(S^k) \coloneqq \underbrace{\frac{1}{M}\sum_{i=1}^{M}\left[\frac{1}{2}(S^k(x^{(i)}))^2 - \log \partial_k S^k(x^{(i)})\right]}_{\text{Empirical Negative Log-Pullback Likelihood } -\frac{1}{M}\sum_i \log \pi_{S^k}(x_k^{(i)}|x_{-k}^{(i)})} + \lambda\Phi(S^k), \tag{14}$$

which is minimized over the space of admissible maps, $\min_{S^k\in\mathcal{S}_k} \mathcal{L}(S^k)$, subject to the strict target monotonicity constraint $\partial_k S^k(x) > 0$ (π-a.e.). The regularization penalty $\Phi(S^k)$ is defined as,

$$\Phi(S^k) \coloneqq \sum_{j=1}^{d}\sqrt{\frac{1}{M}\sum_{i=1}^{M}\left(\frac{\partial S^k(x^{(i)})}{\partial x_j}\right)^2}. \tag{15}$$

1.3 The Generalized Precision Matrix

Spantini, Bigoni, and Marzouk [13] proved that for distributions with twice-differentiable log-densities, pairwise conditional independence is characterized by the vanishing of the mixed second derivatives of the log-density,

$$X_j \perp X_k \mid X_{-jk} \iff \partial_j\partial_k \log \pi(x) = 0 \quad \forall x \in \mathbb{R}^d. \tag{16}$$

Because $\partial_j \partial_k \log \pi(x) = \partial_j \partial_k \log \pi(x_k \mid x_{-k})$, the entries of the *generalized precision matrix* $\Omega \in \mathbb{R}^{d \times d}$ are defined as,

$$\Omega_{jk} \coloneqq \mathbb{E}\left|\partial_j \partial_k \log \pi(x_k \mid x_{-k})\right| = \mathbb{E}\left|\partial_j \partial_k \left[-\frac{1}{2}(S^k(x))^2 + \log \partial_k S^k(x)\right]\right|. \tag{17}$$

Using N independent evaluation samples, the Monte Carlo estimator is,

$$\widehat{\Omega}_{jk} \coloneqq \frac{1}{N}\sum_{i=1}^{N}\left|\partial_j \partial_k \left[-\frac{1}{2}(S^k(x^{(i)}))^2 + \log \partial_k S^k(x^{(i)})\right]\right|. \tag{18}$$

The matrix is symmetrized via $\widehat{\Omega}_{\text{L-SING}} = \frac{1}{2}(\widehat{\Omega} + \widehat{\Omega}^T)$, the diagonal is fixed to $\widehat{\Omega}_{kk} = 1$, and values are normalized to a maximum of 1. Edges are determined by thresholding: $(j,k) \in E \iff |\widehat{\Omega}_{jk}| > \tau$.

[Key Requirement: 2. Higher Regularity (C^3 Smoothness)]

Evaluating the generalized precision entries $\Omega_{jk} = \mathbb{E}|\partial_j \partial_k \log \pi(x_k \mid x_{-k})|$ requires expanding the log-derivative integrand,

$$\begin{aligned}\partial_j \partial_k \left[-\frac{1}{2}(S^k(x))^2 + \log \partial_k S^k(x)\right] &= -(\partial_j S^k)(\partial_k S^k) - S^k(\partial_j \partial_k S^k) \\ &\quad + \frac{(\partial_j \partial_k^2 S^k)(\partial_k S^k) - (\partial_k^2 S^k)(\partial_j \partial_k S^k)}{(\partial_k S^k)^2}.\end{aligned} \tag{19}$$

Consequently, the transport map S^k must possess continuous derivatives up to third order ($S^k \in C^3(\mathbb{R}^d)$, or piecewise C^3) so that the mixed third derivative $\partial_j \partial_k^2 S^k$ is well-defined.

[Key Requirement: 3. Efficient Derivative Computability]

Evaluating (19) across N Monte Carlo evaluation samples and all d^2 pairs (j,k) must be computationally tractable. Under numerical quadrature architectures (such as UMNNs), this requires backpropagating three times through a numerical integrator, creating massive memory footprints and runtime bottlenecks. The transport map should ideally admit closed-form, analytic derivatives to allow direct, instantaneous precision matrix extraction.

2 Limitations of Existing Architectures: MLPs and UMNNs

2.1 The Convexity Trap of Monotonic MLPs

In standard Multi-Layer Perceptrons (MLPs), enforcing monotonicity with respect to an input x_k requires constraining network weights along that input's computational path to be non-negative ($W \geq 0$) and using non-decreasing activation functions (e.g., ReLU or Sigmoids). However, a well-known result by Archer and Wang [1] proves that constraining MLP weights to be non-negative with convex activation functions forces the entire neural network to be **globally convex** [8].

This creates a fundamental bottleneck for transport maps: a convex function cannot model sigmoidal transitions, S-curves, multi-modal density transformations, or higher-order inflections without introducing severe model bias (see Figure 1).

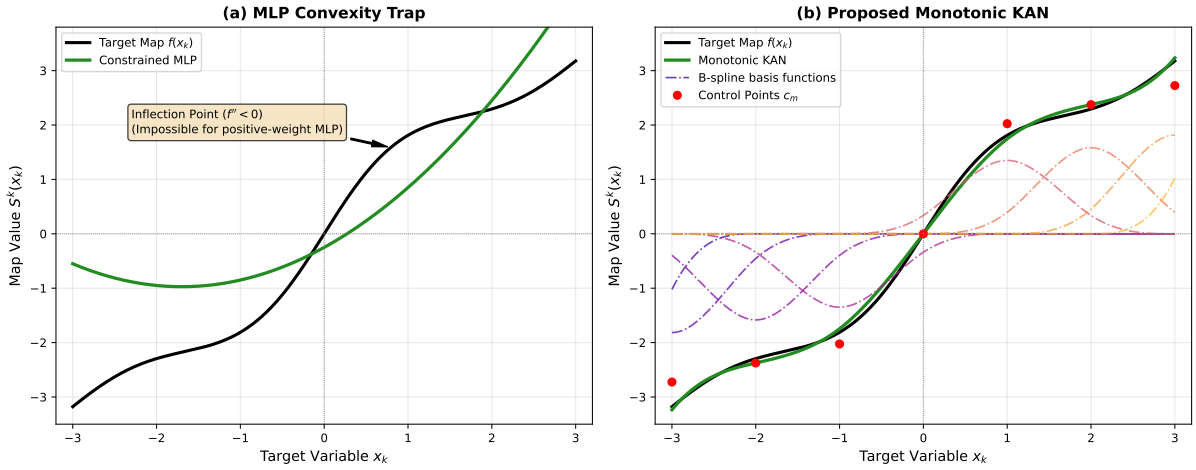


Figure 1: **Function Expressivity and Spline Parameterization. (a)** The MLP convexity trap: positive-weight MLPs [1] are constrained to convex curves ($f'' \geq 0$), failing to fit non-convex inflections of the target map $f(x_k)$. **(b)** Monotonic KAN representation: the learned spline $S^k(x_k)$ (green) fits the target map $f(x_k)$ (black), parameterized by strictly increasing control points c_m (red dots) and the superposition of B-spline basis functions.

2.2 The Quadrature Bottleneck of UMNNs

To circumvent the convexity trap, the baseline L-SING implementation parameterizes $S^k(x)$ using Unconstrained Monotonic Neural Networks (UMNNs) [14],

$$S^k(x) = \int_0^{x_k} f(t, x_{-k}; \psi)\, dt + \beta(x_{-k}), \tag{20}$$

where $f : \mathbb{R}^d \to \mathbb{R}^+$ is an unconstrained neural network strictly constrained to positive outputs via $\exp(\cdot)$ or $\text{softplus}(\cdot)$.

While UMNNs guarantee $\partial_k S^k(x) = f(x; \psi) > 0$ strictly by construction, evaluating $S^k(x)$ requires numerical integration,

$$S^k(x) \approx x_k \sum_{m=1}^{N_{\text{quad}}} w_m f\left(x_k \tau_m, x_{-k}; \psi\right), \tag{21}$$

typically using a 50-node Clenshaw–Curtis quadrature grid per sample. This introduces several major computational drawbacks:

1. **Evaluation Overhead:** Each forward evaluation of $S^k(x)$ requires $N_{\text{quad}} = 50$ forward passes through the network f, increasing training and evaluation runtimes by orders of magnitude.

2. **Nested Autodiff Through Quadrature:** Computing the Rosasco penalty $\Phi(S^k)$ requires taking first derivatives $\partial_j S^k$, requiring backpropagation through the 50-point numerical summation. Computing the generalized precision matrix entries $\widehat{\Omega}_{jk}$ requires third-order automatic differentiation through the numerical quadrature graph, leading to massive GPU/CPU memory consumption and slow extraction times (e.g., $>$ 200 seconds for $d = 40$).

3 Admissibility Requirements for the Transport Map S^k

To replace the UMNN parameterization, we must formalize the exact conditions that an admissible transport map S^k must fulfill.

Definition 3.1 (Transport Map Admissibility Criteria)**.** A parameterization of $S^k : \mathbb{R}^d \to \mathbb{R}$ is admissible for L-SING if and only if it satisfies the following criteria:

1. **Strict Target Positivity / Invertibility (Key Requirement 1):**

$$\partial_k S^k(x) \coloneqq \frac{\partial S^k(x)}{\partial x_k} \geq \epsilon > 0 \quad \forall x \in \mathbb{R}^d. \tag{22}$$

   This ensures that $S^k(x_{-k}, \cdot)$ is a 1D orientation-preserving diffeomorphism on $\mathbb{R}$, guaranteeing the existence of a unique inverse pushforward and preventing numerical divergence $\log \partial_k S^k(x) \to -\infty$.

2. **Global Real-Line Domain ($\mathbb{R}^d$ Extrapolation):** The condition $\partial_k S^k(x) \geq \epsilon > 0$ must hold everywhere on $\mathbb{R}^d$, including outside the compact empirical training domain or spline knot interval $[t_{\min}, t_{\max}]$.

3. **Higher Regularity / C^3 Smoothness (Key Requirement 2):** To evaluate the generalized precision integrand,

$$T_{jk}(x) = \partial_j \partial_k \left[-\frac{1}{2}(S^k(x))^2 + \log \partial_k S^k(x) \right], \tag{23}$$

   S^k must possess continuous derivatives up to order 3 ($S^k \in C^3(\mathbb{R}^d)$, or piecewise continuous 3rd derivatives), ensuring that the mixed derivative $\partial_j \partial_k^2 S^k$ is well-defined.

4. **Conditioning Freedom (Unconstrained on x_{-k}):** For all $j \neq k$, the derivative $\partial_j S^k(x)$ must be **unconstrained in sign**. The conditioning relationship must be capable of learning arbitrary, non-monotonic, multimodal responses to faithfully capture complex conditional distributions $\pi(x_k \mid x_{-k})$.

Remark 3.2 (Mandatory Requirements and Desirable Properties). Beyond these mandatory requirements, an even further nice-to-have is **efficient derivative computability** (Key Requirement 3). In Section 4, we turn to Kolmogorov–Arnold Networks (KANs), establishing a parameterization framework that not only certifies all four mandatory admissibility criteria by construction, but also natively delivers closed-form, analytic derivative evaluations.

4 Theoretical Framework and Monotonicity Proofs for KANs

The architectural foundation of Kolmogorov–Arnold Networks (KANs) [9] is rooted in the Kolmogorov–Arnold representation theorem [5], which establishes that any continuous multivariate function on a compact domain can be expressed as a finite superposition of continuous univariate functions and additions.

Theorem 4.1 (Kolmogorov–Arnold Representation Theorem [5])**.** *Let $f : [0,1]^d \to \mathbb{R}$ be an arbitrary continuous multivariate function. Then f admits the exact representation,*

$$f(x_1, \ldots, x_d) = \sum_{q=1}^{2d+1} \Phi_q \left(\sum_{p=1}^{d} \phi_{q,p}(x_p) \right), \tag{24}$$

where $\Phi_q : \mathbb{R} \to \mathbb{R}$ and $\phi_{q,p} : [0,1] \to \mathbb{R}$ are continuous univariate functions.

While the classical theorem proves representation existence with functions that may be fractal or non-smooth, modern KANs (Liu et al. [9]) operationalize this principle by parameterizing the univariate edge functions with smooth, continuously differentiable splines (such as B-splines) and generalizing the structure to arbitrary network depths $L \geq 2$ and layer widths.

As illustrated in Figure 2, KANs invert the classical MLP design: where MLPs place fixed non-linear activations σ on nodes and learnable linear weights on edges, KANs place learnable 1D spline functions ϕ directly on edges and simple summations $\sum$ at nodes. Crucially, each KAN layer $l \in \{1, \ldots, L\}$ with n_{l-1} inputs and n_l outputs forms a matrix of 1D functions,

$$\mathbf{\Phi}_l = \begin{pmatrix} \phi_{l,1,1}(\cdot) & \cdots & \phi_{l,1,n_{l-1}}(\cdot) \\ \vdots & \ddots & \vdots \\ \phi_{l,n_l,1}(\cdot) & \cdots & \phi_{l,n_l,n_{l-1}}(\cdot) \end{pmatrix}, \tag{25}$$

updating node activations via $x_j^{(l)} = \sum_{i=1}^{n_{l-1}} \phi_{l,j,i}(x_i^{(l-1)})$ from input $x^{(0)} = x$. A deep KAN of depth $L \geq 2$ is then simply the functional composition $\mathrm{KAN}(x) = (\mathbf{\Phi}_L \circ \cdots \circ \mathbf{\Phi}_1)(x)$, directly bridging the classical 2-layer theorem to deep expressive representations.

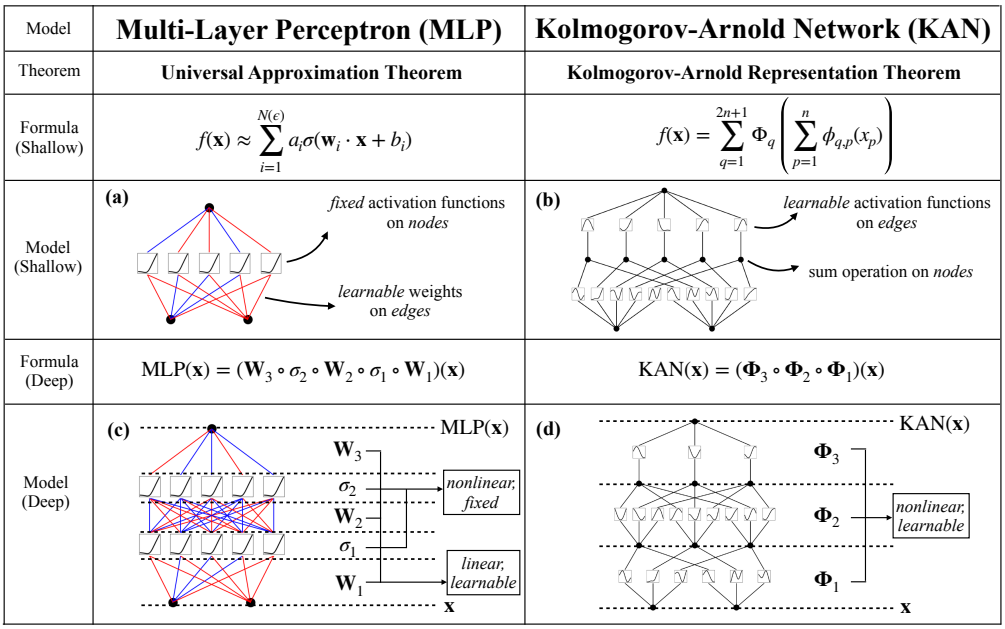


Figure 2: **Architectural Comparison: Multi-Layer Perceptrons (MLPs) vs. Kolmogorov–Arnold Networks (KANs)** (reproduced from Liu et al. [9], Figure 0.1). **(a) MLP Layer:** Fixed non-linear activation functions $\sigma(\cdot)$ (e.g., ReLU, SiLU) reside on nodes (neurons), while learnable scalar linear weights W reside on edges, computing $x^{(l)} = \sigma(W_l x^{(l-1)})$. **(b) KAN Layer:** Learnable activation functions (parameterized as smooth 1D B-splines $\phi(x)$) reside directly on edges, while receiving nodes perform simple linear addition $\sum$ without non-linearities. This 2-layer graph $[d \to 2d+1 \to 1]$ provides the direct computational realization of the classical Kolmogorov–Arnold representation theorem (24). **(c) Deep MLP Architecture:** Stacks multiple layers by interleaving affine weight matrices W and static node non-linearities: $\mathrm{MLP}(x) = (W_L \circ \sigma \circ \cdots \circ W_1)x$. **(d) Deep KAN Architecture:** Generalizes the edge-activation principle to arbitrary depths $L \geq 2$ and layer widths $[n_0, n_1, \ldots, n_L]$. Each layer transition $l \in \{1, \ldots, L\}$ applies an $n_l \times n_{l-1}$ function matrix of 1D splines $\mathbf{\Phi}_l = (\phi_{l,j,i})$ followed by summation $(x_j^{(l)} = \sum_{i=1}^{n_{l-1}} \phi_{l,j,i}(x_i^{(l-1)}))$, defining the deep functional composition $\mathrm{KAN}(x) = (\mathbf{\Phi}_L \circ \cdots \circ \mathbf{\Phi}_1)(x)$.

We now establish that the admissibility requirements of Definition 3.1 can be certified constructively using B-splines. We first isolate the 1D B-spline monotonicity property as a foundational lemma, demonstrate why additive ($L = 1$) splines are structurally insufficient for conditional independence discovery, and then analyze the canonical two-layer ($L = 2$) KAN before generalizing to arbitrary network depths.

4.1 Foundational Lemma: Monotonic 1D B-Splines

The fundamental mathematical building block across our monotonic KAN architectures is the univariate B-spline with ordered control points. Crucially, enforcing monotonicity at discrete control points guarantees monotonicity of the entire continuous spline curve.

Lemma 4.2 (Monotonicity of 1D B-Splines)**.** *Let $\phi(t) = \sum_{m=1}^{M} c_m B_{m,p}(t)$ be a univariate B-spline of degree $p \geq 1$ defined over a non-decreasing knot sequence $t_1 \leq t_2 \leq \cdots \leq t_{M+p+1}$ with $t_{m+p} > t_m$ for all $m \in \{2, \ldots, M\}$. If the control coefficients are non-decreasing ($c_{m+1} - c_m \geq 0$), then $\phi'(t) \geq 0$ everywhere on the internal active domain $[t_{p+1}, t_{M+1}]$. Moreover, if the control coefficients satisfy strictly positive increments,*

$$c_{m+1} - c_m \geq \delta > 0 \quad \forall m \in \{1, \ldots, M-1\}, \tag{26}$$

then $\phi'(t) \geq \epsilon > 0$ everywhere on $[t_{p+1}, t_{M+1}]$. Combined with C^1-continuous linear tail extrapolation, both derivative bounds ($\phi'(t) \geq 0$ and $\phi'(t) \geq \epsilon > 0$, respectively) hold globally for all $t \in \mathbb{R}$.

Proof. By the standard derivative recurrence for B-splines of degree p,

$$\phi'(t) = p \sum_{m=2}^{M} \frac{c_m - c_{m-1}}{t_{m+p} - t_m} B_{m,p-1}(t). \tag{27}$$

Because the basis functions of degree $p-1$ satisfy $B_{m,p-1}(t) \geq 0$ and form a partition of unity on $[t_{p+1}, t_{M+1}]$ with positive knot spans $t_{m+p} - t_m > 0$, $\phi'(t)$ is a convex combination of scaled coefficient differences. Whenever $c_m - c_{m-1} \geq 0$, every term in (27) is non-negative, yielding $\phi'(t) \geq 0$. Under the strict increment condition $c_m - c_{m-1} \geq \delta > 0$, $\phi'(t)$ is a convex combination of strictly positive quantities,

$$\phi'(t) \geq \min_m \left(\frac{p\,\delta}{t_{m+p} - t_m} \right) \coloneqq \epsilon > 0. \tag{28}$$

With linear tail extrapolation maintaining constant boundary slopes ($\phi'(t) = \phi'(t_{\text{boundary}})$) outside the active knot interval (Section 5.2), non-negativity and strict positivity extend over the entire real line $\mathbb{R}$. □

Remark 4.3 (Why Additive Splines ($L = 1$) Cannot Discover General Graph Structure). Consider a purely additive single-layer spline transport map $S^k(x) = \phi_k(x_k) + \sum_{j \neq k} \phi_j(x_j)$. Applying Lemma 4.2 to ϕ_k immediately guarantees $\partial_k S^k(x) = \phi_k'(x_k) \geq \epsilon > 0$.

However, because the conditioning coordinates enter purely additively, the target derivative $\partial_k S^k(x) = \phi_k'(x_k)$ is completely independent of x_{-k}, so all mixed cross-derivatives of the map vanish identically,

$$\partial_j \partial_k S^k(x) = \frac{\partial}{\partial x_j}[\phi_k'(x_k)] \equiv 0 \quad \text{and} \quad \partial_j \partial_k \log \partial_k S^k(x) \equiv 0 \quad \forall j \neq k. \tag{29}$$

In the generalized precision expansion (19), both the interaction term $-S^k(\partial_j \partial_k S^k)$ and the log-Jacobian coupling term vanish, reducing the integrand to the separable rank-one product $-(\partial_j S^k)(\partial_k S^k) = -\phi_j'(x_j)\phi_k'(x_k)$. Geometrically, x_{-k} can only induce a rigid location shift without modulating conditional scale, variance, or higher-order shape; for any zero-mean scale or shape coupling, $\phi_j \equiv 0$ and Ω_{jk} in (17) collapses to zero. Functional composition ($L \geq 2$) is therefore mathematically necessary to achieve $\partial_j \partial_k S^k(x) \not\equiv 0$.

4.2 Two-Layer Monotonic KAN as the Primary Workhorse

Having established that additive splines fail the graphical model recovery requirement, we focus on the canonical two-layer KAN $[d \to Q \to 1]$ illustrated in Figure 3. This architecture directly operationalizes the classical Kolmogorov–Arnold representation theorem (Theorem 4.1) and serves as the primary parameterization for local transport maps S^k.

Definition 4.4 (Two-Layer Monotonic KAN)**.** A 2-layer KAN transport map $S^k : \mathbb{R}^d \to \mathbb{R}$ $(L = 2)$ with $n_1 = Q$ hidden summation nodes and input $x^{(0)} = x$ is parameterized via,

$$S^k(x) = x_1^{(2)} = \sum_{q=1}^{Q} \phi_{2,1,q}\big(x_q^{(1)}(x)\big), \tag{30}$$

where $x_q^{(1)}(x)$ denotes the q-th hidden activation at layer $l = 1$,

$$x_q^{(1)}(x) \coloneqq \phi_{1,q,k}(x_k) + \sum_{j \neq k} \phi_{1,q,j}(x_j), \tag{31}$$

and the edge activations satisfy:

1. **Target Inner Splines ($l = 1$):** $\phi_{1,q,k} : \mathbb{R} \to \mathbb{R}$ are non-decreasing univariate B-splines $(\phi'_{1,q,k} \geq 0)$, parameterized via Lemma 4.2.
2. **Outer Splines ($l = 2$):** $\phi_{2,1,q} : \mathbb{R} \to \mathbb{R}$ are non-decreasing univariate B-splines $(\phi'_{2,1,q} \geq 0)$, with at least one active path $q^* \in \{1, \ldots, Q\}$ satisfying $\phi'_{1,q^*,k} \geq \epsilon_1 > 0$ and $\phi'_{2,1,q^*} \geq \epsilon_2 > 0$, parameterized via Lemma 4.2.
3. **Conditioning Inner Splines ($l = 1$):** $\phi_{1,q,j} : \mathbb{R} \to \mathbb{R}$ $(j \neq k)$ are arbitrary unconstrained univariate B-splines.

Theorem 4.5 (Admissibility and Structural Coupling in 2-Layer KANs)**.** *Let S^k be parameterized according to Definition 4.4. Then,*

1. ***Target Monotonicity (Criterion 1):*** *For all $x \in \mathbb{R}^d$, the target derivative satisfies,*

$$\partial_k S^k(x) = \sum_{q=1}^{Q} \phi'_{2,1,q}\big(x_q^{(1)}(x)\big) \cdot \phi'_{1,q,k}(x_k) \geq \epsilon > 0, \tag{32}$$

   where $\epsilon \coloneqq \epsilon_1 \epsilon_2 > 0$ is a uniform positive constant, certifying that $S^k(x_{-k}, \cdot)$ is a 1D orientation-preserving diffeomorphism on $\mathbb{R}$.

2. ***Conditioning Freedom and Structural Coupling (Criterion 4):*** *For any conditioning coordinate $j \neq k$, the conditioning splines $\phi_{1,q,j}$ are unconstrained in sign, and the mixed second derivative is given analytically by,*

$$\partial_j \partial_k S^k(x) = \sum_{q=1}^{Q} \phi''_{2,1,q}\big(x_q^{(1)}(x)\big) \cdot \phi'_{1,q,j}(x_j) \cdot \phi'_{1,q,k}(x_k). \tag{33}$$

   Consequently, $\partial_j \partial_k S^k(x) \not\equiv 0$ in general whenever the outer splines exhibit non-zero curvature, allowing unconstrained modeling of conditional dependencies and full recovery of the generalized precision matrix Ω.

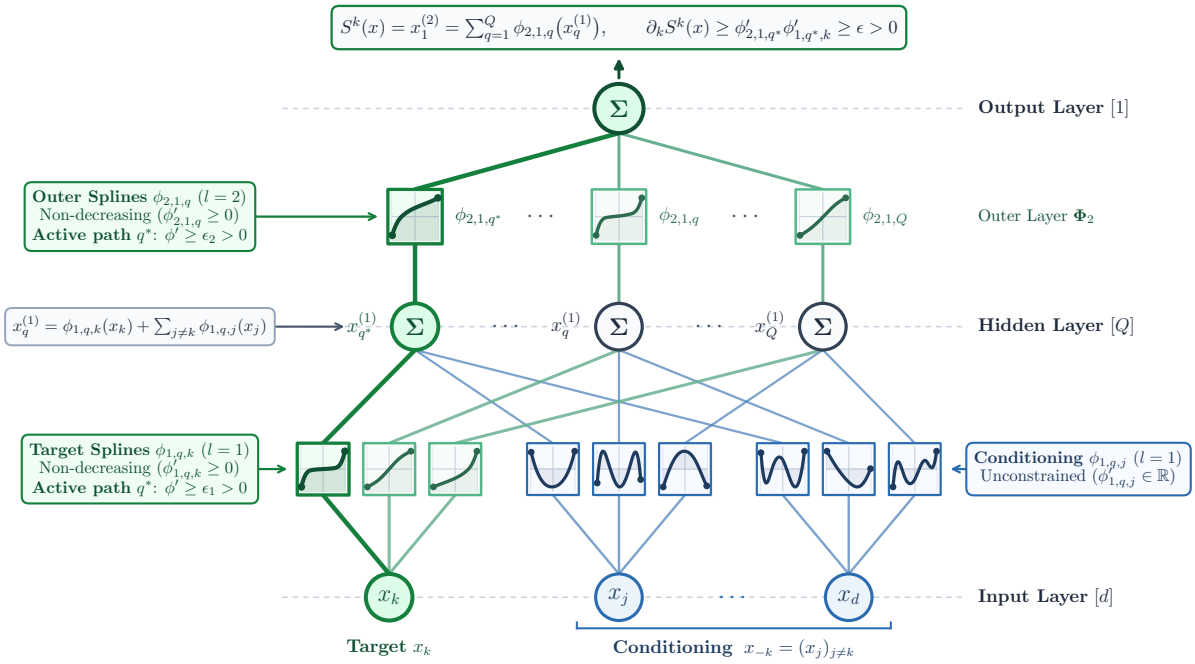


Figure 3: **Two-Layer Monotonic KAN Transport Map $S^k : \mathbb{R}^d \to \mathbb{R}$ with Topology** $[d \to Q \to 1]$ (following the edge-activation diagram convention of Liu et al. [9]). Input coordinates $x^{(0)} = x$ partition into the target feature x_k (green) and conditioning features $x_{-k} = (x_j)_{j \neq k}$ (blue), while hidden nodes $x_q^{(1)}$ and the output node $x_1^{(2)} = S^k(x)$ perform linear summations ($\sum$). Each edge carries a learnable 1D B-spline plot: target inner splines $\phi_{1,q,k}(x_k)$ and outer splines $\phi_{2,1,q}(x_q^{(1)})$ are non-decreasing ($\phi'_{1,q,k} \geq 0$, $\phi'_{2,1,q} \geq 0$, muted green), and the certified active path through hidden node $x_{q^*}^{(1)}$ (highlighted in bold dark green) maintains strictly positive derivatives $\phi'_{1,q^*,k} \geq \epsilon_1 > 0$ and $\phi'_{2,1,q^*} \geq \epsilon_2 > 0$, guaranteeing $\partial_k S^k(x) \geq \epsilon_1 \epsilon_2 > 0$ globally on $\mathbb{R}^d$, whereas conditioning inner splines $\phi_{1,q,j}(x_j)$ remain unconstrained (blue).

Proof. Differentiating $S^k(x)$ in (30) with respect to x_k via the multivariable chain rule yields,

$$\partial_k S^k(x) = \sum_{q=1}^{Q} \frac{\partial \phi_{2,1,q}(x_q^{(1)})}{\partial x_q^{(1)}} \cdot \frac{\partial x_q^{(1)}(x)}{\partial x_k} = \sum_{q=1}^{Q} \phi'_{2,1,q}\big(x_q^{(1)}(x)\big) \cdot \phi'_{1,q,k}(x_k). \tag{34}$$

By Lemma 4.2, all target inner and outer splines satisfy $\phi'_{1,q,k} \geq 0$ and $\phi'_{2,1,q} \geq 0$ globally on $\mathbb{R}$, ensuring every product term in the sum is non-negative. For the designated active path q^*, the product satisfies $\phi'_{2,1,q^*}(x_{q^*}^{(1)}(x)) \cdot \phi'_{1,q^*,k}(x_k) \geq \epsilon_1\epsilon_2 =: \epsilon > 0$. Decomposing the summation into the active path q^* and the remaining non-negative paths yields $\partial_k S^k(x) \geq \epsilon + 0 = \epsilon > 0$ for all $x \in \mathbb{R}^d$.

Next, differentiating $\partial_k S^k(x)$ with respect to conditioning variable x_j ($j \neq k$) via the chain rule pulls out the inner derivative $\partial_j x_q^{(1)}(x) = \phi'_{1,q,j}(x_j)$, yielding (33). Because $\phi_{1,q,j}$ is unconstrained, $\phi'_{1,q,j}(x_j)$ can take arbitrary real values, and the second derivative of the outer spline $\phi''_{2,1,q}(x_q^{(1)}(x))$ is likewise unconstrained in sign while maintaining $\phi'_{2,1,q} \geq 0$. Thus, $\partial_j \partial_k S^k(x)$ is generally non-zero, allowing the map to capture non-Gaussian dependency structures. □

Remark 4.6 (Remaining Criteria: Domain Extrapolation, Regularity, and Analytic Differentiation). The remaining criteria of Definition 3.1 follow naturally from the spline construction: **Global Extrapolation (Criterion 2)** is satisfied by attaching C^1 linear tails outside the knot domain (Section 5.2); C^3 **Smoothness (Criterion 3)** is satisfied by selecting B-spline degree $p = 3$ for piecewise continuous third derivatives (C^2 across interior knots) or $p \geq 4$ for continuous third derivatives on the interior knot domain; and **Derivative Computability (Key Requirement 3)** is fulfilled natively because all derivatives in (32) and (33) are available in closed form via standard B-spline recurrence, bypassing the nested numerical quadrature required by UMNNs.

4.3 Arbitrarily Deep Multi-Layer KAN (L Layers)

We now generalize the construction to a KAN of arbitrary depth $L \geq 2$, with layer dimensions $[n_0, n_1, n_2, \ldots, n_L]$, where $n_0 = d$ (input features) and $n_L = 1$ (scalar output $S^k(x)$), as illustrated in Figure 4.

Definition 4.7 (Arbitrarily Deep Monotonic KAN)**.** Let $S^k : \mathbb{R}^d \to \mathbb{R}$ be an L-layer KAN ($L \geq 2$) with layer dimensions $[n_0, n_1, \ldots, n_L]$, where $n_0 = d$ (input features) and $n_L = 1$ (scalar transport map output). Setting the input layer to $x^{(0)} = x \in \mathbb{R}^d$, the network computes intermediate activations layer-by-layer for $l \in \{1, \ldots, L\}$,

$$x_j^{(l)} = \sum_{i=1}^{n_{l-1}} \phi_{l,j,i}\big(x_i^{(l-1)}\big), \quad j = 1, \ldots, n_l, \tag{35}$$

where $\phi_{l,j,i} : \mathbb{R} \to \mathbb{R}$ is the learnable univariate B-spline on the edge from neuron i in layer $l-1$ to neuron j in layer l. The full transport map $S^k(x) := x_1^{(L)}$ represents the L-fold functional composition,

$$S^k(x) = \Big(\mathbf{\Phi}_L \circ \mathbf{\Phi}_{L-1} \circ \cdots \circ \mathbf{\Phi}_1\Big)(x). \tag{36}$$

Let $\mathbb{P}_k$ denote the set of all directed computational paths $\mathcal{P} = (i_0, i_1, \ldots, i_L)$ from target input $i_0 = k$ to output $i_L = 1$, where $i_l \in \{1, \ldots, n_l\}$. The network is an *admissible monotonic KAN* if its edge splines, parameterized via Lemma 4.2, satisfy:

1. **Target-Transmitting Splines:** Every edge emanating from target input x_k at layer $l = 1$ and all downstream edges across hidden and output layers ($2 \leq l \leq L$) are non-decreasing,

$$\phi'_{1,j,k}(t) \geq 0 \quad (\forall j \in \{1, \ldots, n_1\}), \quad \text{and} \quad \phi'_{l,j,i}(t) \geq 0 \quad (\forall 2 \leq l \leq L, \forall j, i), \quad \forall t \in \mathbb{R}, \tag{37}$$

and there exists at least one active backbone path $\mathcal{P}^* = (i_0^*, i_1^*, \ldots, i_L^*) \in \mathbb{P}_k$ (with $i_0^* = k$ and $i_L^* = 1$) whose edges are strictly increasing,

$$\phi'_{l,i_l^*,i_{l-1}^*}(t) \geq \epsilon_l > 0 \quad \forall t \in \mathbb{R}, \quad \forall l \in \{1, \ldots, L\}. \tag{38}$$

2. **Conditioning Splines:** For all conditioning inputs $i \neq k$ at layer $l = 1$, the splines $\phi_{1,j,i}$ are unconstrained in sign, derivative, and magnitude.

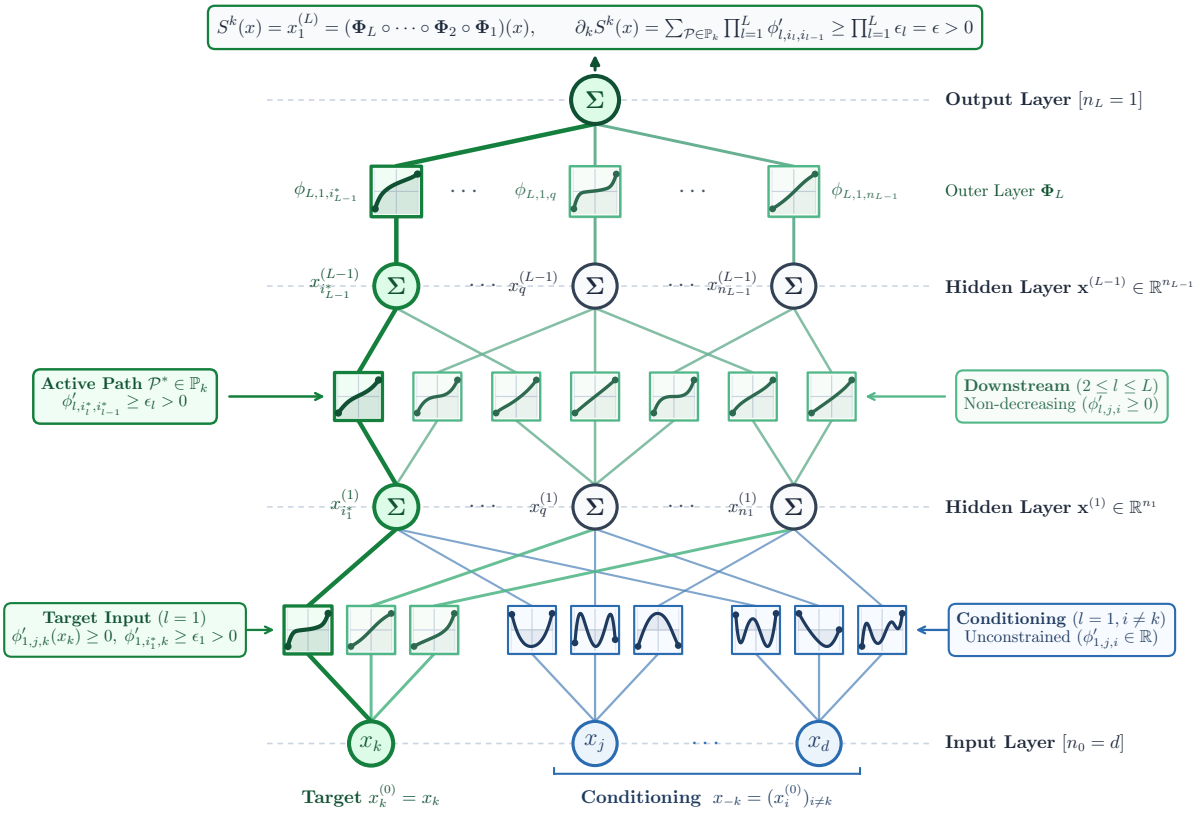


Figure 4: **Arbitrarily Deep L-Layer Monotonic KAN Transport Map** $[n_0 \to n_1 \to \cdots \to n_{L-1} \to n_L]$ ($n_0 = d, n_L = 1$; following Liu et al. [9], Figure 0.1d). Target edges emanating from x_k at layer $l = 1$ and all downstream hidden edges ($2 \leq l \leq L$) are non-decreasing ($\phi'_{l,j,i} \geq 0$, muted green), while conditioning edges $\phi_{1,j,i}(x_i)$ ($i \neq k$) are unconstrained (blue). At least one certified active backbone path $\mathcal{P}^* = (i_0^*, i_1^*, \ldots, i_{L-1}^*, i_L^*) \in \mathbb{P}_k$ with $i_0^* = k, i_L^* = 1$ (highlighted in bold dark green) maintains strictly positive derivatives $\phi'_{l,i_l^*,i_{l-1}^*} \geq \epsilon_l > 0$, guaranteeing $\partial_k S^k(x) \geq \prod_{l=1}^L \epsilon_l = \epsilon > 0$ globally on $\mathbb{R}^d$.

Theorem 4.8 (Admissibility and Certified Monotonicity for Arbitrarily Deep KANs)**.** *Let S^k be parameterized according to Definition 4.7. Then,*

1. ***Target Monotonicity (Criterion 1):*** *For all $x \in \mathbb{R}^d$, the target derivative satisfies,*

$$\partial_k S^k(x) = \sum_{\mathcal{P}\in\mathbb{P}_k} \prod_{l=1}^{L} \phi'_{l,i_l,i_{l-1}}\big(x_{i_{l-1}}^{(l-1)}\big) \geq \epsilon > 0, \tag{39}$$

where $\epsilon := \prod_{l=1}^L \epsilon_l > 0$ is a uniform positive constant, certifying that $S^k(x_{-k}, \cdot)$ is a 1D orientation-preserving diffeomorphism on $\mathbb{R}$.

2. ***Conditioning Freedom (Criterion 4):*** *For all conditioning coordinates $i \neq k$, the conditioning splines $\phi_{1,j,i}$ are unconstrained in sign, derivative, and magnitude, allowing unrestricted nonlinear contextual modulation while strictly preserving target monotonicity.*

Proof. We evaluate the sensitivity of the transport map output $S^k(x) = x_1^{(L)}$ with respect to the target input $x_k = x_k^{(0)}$. Applying the multivariable chain rule layer by layer from the output layer L back to the input layer 0 expands the total derivative into a multi-index sum,

$$\frac{\partial S^k(x)}{\partial x_k} = \frac{\partial x_1^{(L)}}{\partial x_k^{(0)}} = \sum_{i_{L-1}=1}^{n_{L-1}} \sum_{i_{L-2}=1}^{n_{L-2}} \cdots \sum_{i_1=1}^{n_1} \left(\frac{\partial x_1^{(L)}}{\partial x_{i_{L-1}}^{(L-1)}} \frac{\partial x_{i_{L-1}}^{(L-1)}}{\partial x_{i_{L-2}}^{(L-2)}} \cdots \frac{\partial x_{i_1}^{(1)}}{\partial x_k^{(0)}} \right). \tag{40}$$

From the layer definition (35), intermediate neuron activations are linear superpositions of univariate edge splines, $x_j^{(l)} = \sum_{i=1}^{n_{l-1}} \phi_{l,j,i}(x_i^{(l-1)})$. Differentiating $x_j^{(l)}$ with respect to upstream activation $x_i^{(l-1)}$, all other terms vanish, yielding the univariate spline derivative,

$$\frac{\partial x_j^{(l)}}{\partial x_i^{(l-1)}} = \phi'_{l,j,i}\big(x_i^{(l-1)}\big). \tag{41}$$

At the first layer $l = 1$, holding all conditioning coordinates $x_{-k} = \{x_i : i \neq k\}$ fixed gives $\partial x_{i_1}^{(1)} / \partial x_k^{(0)} = \phi'_{1,i_1,k}(x_k^{(0)})$.

Substituting these edge derivatives into (40), each term in the summation corresponds to the product of edge derivatives along a directed computational path $\mathcal{P} = (i_0, i_1, \ldots, i_L) \in \mathbb{P}_k$ with $i_0 = k$ and $i_L = 1$,

$$\partial_k S^k(x) = \sum_{\mathcal{P} \in \mathbb{P}_k} \prod_{l=1}^{L} \phi'_{l,i_l,i_{l-1}}\big(x_{i_{l-1}}^{(l-1)}\big). \tag{42}$$

By Lemma 4.2 and Condition 1 of Definition 4.7, all target-transmitting splines satisfy $\phi'_{l,j,i}(t) \geq 0$ globally on $\mathbb{R}$, so every path product in (42) is non-negative regardless of intermediate activations $x^{(l-1)}$. Isolating the certified active backbone path $\mathcal{P}^* = (i_0^*, i_1^*, \ldots, i_L^*)$, along which $\phi'_{l,i_l^*,i_{l-1}^*}(t) \geq \epsilon_l > 0$, we decompose the sum as,

$$\partial_k S^k(x) = \prod_{l=1}^{L} \phi'_{l,i_l^*,i_{l-1}^*}\big(x_{i_{l-1}^*}^{(l-1)}\big) + \sum_{\mathcal{P} \neq \mathcal{P}^*} \prod_{l=1}^{L} \phi'_{l,i_l,i_{l-1}}\big(x_{i_{l-1}}^{(l-1)}\big) \geq \prod_{l=1}^{L} \epsilon_l + 0 = \epsilon > 0. \tag{43}$$

Because this strictly positive bound holds independently of intermediate activations $x^{(l-1)}$ and does not depend on x, $S^k(x)$ is certified strictly monotonic across all of $\mathbb{R}^d$, establishing Criterion 1.

Finally, conditioning coordinates x_{-k} enter only through conditioning splines $\phi_{1,j,i}$ $(i \neq k)$ at layer $l = 1$. Because these splines are completely unconstrained (Condition 2 of Definition 4.7), they provide unrestricted non-linear modulation of intermediate activations without altering the non-negativity of downstream derivatives, establishing Criterion 4. □

4.4 Scale-Modulated Monotonic KAN (SM-KAN)

4.4.1 The Additive Polarization Dilemma

In non-Gaussian graphical models, conditional dependencies are frequently *multiplicative* or *scale-dependent* (heteroscedastic). Consider, for example, the benchmark Butterfly distribution,

$$X_1 \sim \mathcal{N}(0,1), \quad W \sim \mathcal{N}(0,1), \quad X_1 \perp W, \quad X_2 = W \cdot X_1. \tag{44}$$

Conditioned on $X_1 = x_1 \neq 0$, the target variable follows $X_2 \mid (X_1 = x_1) \sim \mathcal{N}(0, x_1^2)$, so the exact Knothe–Rosenblatt transport map normalizing X_2 to a standard Gaussian is,

$$S^2(x_1, x_2) = \frac{x_2}{|x_1|}, \quad x_1 \neq 0. \tag{45}$$

In a purely additive 2-layer KAN (30), each hidden node combines the target spline $u_q = \phi_{1,q,k}(x_k)$ and the conditioning sum $v_q = \sum_{j \neq k} \phi_{1,q,j}(x_j)$ solely by addition, $x_q^{(1)}(x) = u_q + v_q$. Representing a multiplicative interaction $u \cdot v$ through univariate outer functions applied to sums $u \pm v$ requires algebraic polarization,

$$u \cdot v = \frac{1}{4}(u+v)^2 - \frac{1}{4}(u-v)^2. \tag{46}$$

When we expand $(u+v)^2 = u^2 + 2uv + v^2$, isolating the cross-product $u \cdot v$ requires subtracting a second branch $-\frac{1}{4}(u-v)^2$ to cancel the unwanted marginal squares u^2 and v^2. In a standard unconstrained KAN, this subtraction is achieved by assigning negative outer spline derivatives ($\phi'_{2,1,q} < 0$) to the second branch. Under Definition 4.4, however, global target monotonicity requires $\phi'_{2,1,q}(t) \geq 0$ across all outer splines to guarantee $\partial_k S^k(x) > 0$ (and furthermore, the quadratic outer functions $t \mapsto \pm\frac{1}{4}t^2$ themselves are not monotonic on $\mathbb{R}$ since their derivatives $\pm\frac{1}{2}t$ change sign at $t = 0$). Consequently, purely additive monotonic KANs cannot represent multiplicative conditioning without violating the non-negative outer-slope constraints required for target monotonicity.

4.4.2 Scale-Modulation Architecture

The polarization conflict in (46) arises only when a purely additive network is forced to synthesize multiplication indirectly by squaring and subtracting sums $u \pm v$. To resolve this conflict directly, we introduce an explicit multiplicative conditioning pathway outside the core monotonic KAN: we multiply a certified monotonic core network $S^k_{\text{core}}(x)$ by a strictly positive conditioning scale factor $E(\sigma(x_{-k}))$ and add a conditioning location shift $\alpha(x_{-k})$.

Here $\sigma(x_{-k}) = \sum_{j \neq k} \psi_j(x_j)$ and $\alpha(x_{-k}) = \sum_{j \neq k} \rho_j(x_j)$ are unconstrained additive B-spline networks over the conditioning variables x_{-k} (which may also be generalized to multi-layer KANs). Because $\sigma(x_{-k})$ and $\alpha(x_{-k})$ depend only on the conditioning coordinates x_{-k}, the product $E(\sigma(x_{-k})) \cdot S^k_{\text{core}}(x)$ is formed directly at the output without adding x_k and x_{-k} inside a quadratic outer spline. Consequently, no unwanted marginal squares u^2 or v^2 are generated, and no negative outer spline slopes ($\phi'_{2,1,q} < 0$) are required to cancel them.

To ensure that the scale multiplier remains strictly positive and uniformly bounded away from zero and infinity across all of $\mathbb{R}^{d-1}$ without numerical overflow or derivative collapse in the far tails, the scalar output $\sigma(x_{-k}) \in \mathbb{R}$ is mapped to the positive scale factor through the smooth saturated exponential activation,

$$E(\sigma) \coloneqq \exp\left(s_{\max} \tanh\left(\frac{\sigma}{s_{\max}}\right)\right) \in \left(e^{-s_{\max}}, e^{s_{\max}}\right), \tag{47}$$

where $s_{\max} > 0$ is a fixed scale ceiling (in our implementation, $s_{\max} = 4.0$). Near the origin ($\sigma \approx 0$), $s_{\max} \tanh(\sigma / s_{\max}) \approx \sigma$, so (47) reproduces the standard exponential scale factor $\exp(\sigma)$, while globally guaranteeing $e^{-s_{\max}} < E(\sigma) < e^{s_{\max}}$ and preserving C^∞ smoothness everywhere on $\mathbb{R}$.

Placing the additive offset $\alpha(x_{-k})$ outside the product—directly analogous to the conditioning intercept $+\beta(x_{-k})$ added after the monotone integral in the UMNN parameterization (20)—completes a full location–scale conditioning wrapper around $S^k_{\text{core}}(x)$. Whereas any conditioning shift inside $S^k_{\text{core}}(x)$ is multiplied by $E(\sigma(x_{-k}))$, the external offset $\alpha(x_{-k})$ shifts the standardized output $z_k = S^k(x)$ directly in the $\mathcal{N}(0,1)$ reference space without being attenuated or amplified

by the scale factor. For a general heteroscedastic Butterfly distribution with conditional mean shift $\mu(X_1)$, where $X_2 \mid (X_1 = x_1) \sim \mathcal{N}(\mu(x_1), x_1^2)$, writing the exact transport map as,

$$S^2(x_1, x_2) = \frac{x_2 - \mu(x_1)}{|x_1|} = \exp\big(-\log|x_1|\big) \cdot x_2 - \frac{\mu(x_1)}{|x_1|}, \quad x_1 \neq 0, \tag{48}$$

allows the core monotonic network to learn the linear identity $S^2_{\text{core}}(x_1, x_2) = x_2$ ($\partial_2 S^2_{\text{core}} = 1 > 0$), while $\sigma(x_1) = \psi_1(x_1)$ approximates $-\log|x_1|$ and $\alpha(x_1) = \rho_1(x_1)$ captures the location shift $-\mu(x_1)/|x_1|$ (recovering the centered Butterfly map (45) when $\mu(x_1) \equiv 0$; see Appendix A for a detailed derivation).

Definition 4.9 (Scale-Modulated Monotonic KAN)**.** A **Scale-Modulated Monotonic KAN (SM-KAN)** parameterizes the k-th conditional transport map $S^k : \mathbb{R}^d \to \mathbb{R}$ as,

$$S^k(x) = \exp\left(s_{\max} \tanh\left(\frac{\sigma(x_{-k})}{s_{\max}}\right)\right) \cdot S^k_{\text{core}}(x) + \alpha(x_{-k}), \tag{49}$$

where $s_{\max} > 0$ is a fixed scale ceiling, and:

1. **Certified Monotonic Core (S^k_{core}):** $S^k_{\text{core}} : \mathbb{R}^d \to \mathbb{R}$ is an L-layer certified monotonic KAN ($L \geq 2$, parameterized according to Definition 4.4 or Definition 4.7) satisfying $\partial_k S^k_{\text{core}}(x) \geq \epsilon > 0$ for all $x \in \mathbb{R}^d$.

2. **Conditioning Scale Network (σ):** $\sigma : \mathbb{R}^{d-1} \to \mathbb{R}$ is an unconstrained additive B-spline network over the conditioning coordinates x_{-k},

$$\sigma(x_{-k}) = \sum_{j \neq k} \psi_j(x_j), \tag{50}$$

   where each $\psi_j : \mathbb{R} \to \mathbb{R}$ ($j \neq k$) is an unconstrained univariate B-spline.

3. **Conditioning Offset Network (α):** $\alpha : \mathbb{R}^{d-1} \to \mathbb{R}$ is an unconstrained additive B-spline network over the conditioning coordinates x_{-k},

$$\alpha(x_{-k}) = \sum_{j \neq k} \rho_j(x_j), \tag{51}$$

   where each $\rho_j : \mathbb{R} \to \mathbb{R}$ ($j \neq k$) is an unconstrained univariate B-spline.

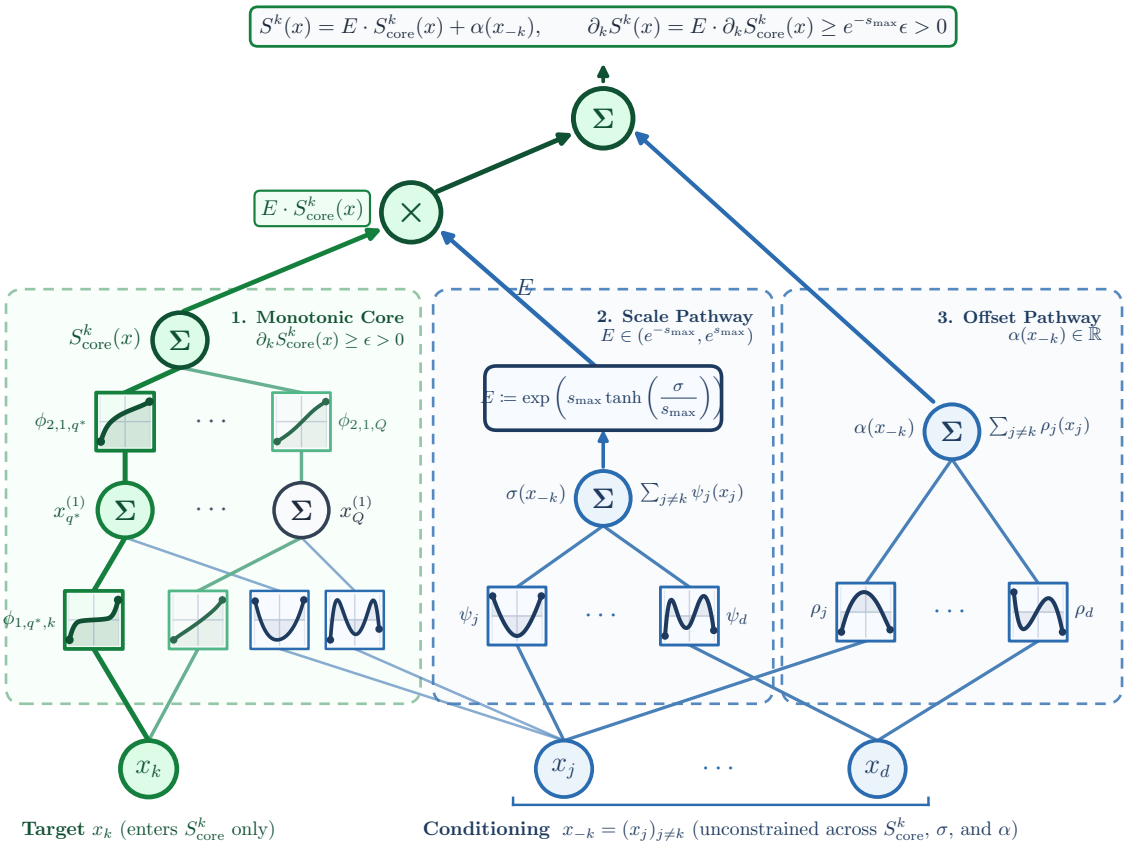


Figure 5: **Scale-Modulated Monotonic KAN (SM-KAN) Transport Map Architecture** (Definition 4.9). The target coordinate x_k (green) enters exclusively through the certified monotonic core network $S^k_{\text{core}}(x)$ (Module 1, left), where non-decreasing target-transmitting splines and an active path q^* guarantee $\partial_k S^k_{\text{core}}(x) \geq \epsilon > 0$. Conditioning coordinates $x_{-k} = (x_j)_{j\neq k}$ (blue) modulate the map through three unconstrained pathways: internal conditioning splines within $S^k_{\text{core}}(x)$, an additive scale network $\sigma(x_{-k}) = \sum_{j\neq k} \psi_j(x_j)$ mapped through smooth saturation and exponentiation $E \coloneqq \exp(s_{\max} \tanh(\sigma/s_{\max})) \in (e^{-s_{\max}}, e^{s_{\max}})$ to a multiplication gate ($\times$) computing $E \cdot S^k_{\text{core}}(x)$ (Module 2, center), and an additive offset network $\alpha(x_{-k}) = \sum_{j\neq k} \rho_j(x_j)$ summed at the output node ($\sum$) (Module 3, right).

Proposition 4.10 (Admissibility and Monotonicity Invariance Under Scale Modulation)**.** *Let S^k be parameterized according to Definition 4.9, and let $E(u) \coloneqq \exp(s_{\max} \tanh(u/s_{\max}))$. Then,*

1. ***Uniform Target Monotonicity (Criterion 1):*** *For all $x \in \mathbb{R}^d$, the target derivative satisfies,*
$$\partial_k S^k(x) = E\big(\sigma(x_{-k})\big) \cdot \partial_k S^k_{core}(x) \geq e^{-s_{\max}} \cdot \epsilon =: \epsilon_{SM} > 0, \tag{52}$$
certifying that for every fixed $x_{-k} \in \mathbb{R}^{d-1}$, the fiber map $S^k(x_{-k}, \cdot)$ is a 1D orientation-preserving diffeomorphism on $\mathbb{R}$ with uniform positive derivative lower bound $\epsilon_{SM} > 0$.

2. ***Conditioning Freedom and Multiplicative Coupling (Criterion 4):*** *For any conditioning coordinate $j \neq k$, the scale and offset splines ψ_j, ρ_j are unconstrained in sign, and the mixed second derivative is given by,*
$$\partial_j \partial_k S^k(x) = E\big(\sigma(x_{-k})\big) \left[\operatorname{sech}^2\left(\frac{\sigma(x_{-k})}{s_{\max}}\right) \psi_j'(x_j) \cdot \partial_k S^k_{core}(x) + \partial_j \partial_k S^k_{core}(x)\right]. \tag{53}$$

In particular, $\partial_j \partial_k S^k(x)$ is non-zero whenever $\psi_j'(x_j) \neq 0$, enabling unrestricted multiplicative conditioning without violating target monotonicity.

Proof. Because $\sigma(x_{-k})$ and $\alpha(x_{-k})$ depend only on conditioning coordinates x_{-k}, they are constant with respect to target coordinate x_k, so $\partial_k[E(\sigma(x_{-k}))] \equiv 0$ and $\partial_k \alpha(x_{-k}) \equiv 0$. Differentiating (49) with respect to x_k yields,

$$\partial_k S^k(x) = E\big(\sigma(x_{-k})\big) \cdot \partial_k S^k_{\text{core}}(x). \tag{54}$$

Since $\tanh(u) > -1$ for all $u \in \mathbb{R}$, we have $E(\sigma(x_{-k})) > e^{-s_{\max}}$. Combined with $\partial_k S^k_{\text{core}}(x) \geq \epsilon > 0$ from Condition 1 of Definition 4.9, this yields the uniform bound $\partial_k S^k(x) \geq e^{-s_{\max}}\epsilon =: \epsilon_{\text{SM}} > 0$ across all $x \in \mathbb{R}^d$, establishing Criterion 1.

Differentiating $\partial_k S^k(x)$ with respect to conditioning coordinate x_j $(j \neq k)$ via the product rule gives $\partial_j \partial_k S^k(x) = \partial_j[E(\sigma(x_{-k}))] \cdot \partial_k S^k_{\text{core}}(x) + E(\sigma(x_{-k})) \cdot \partial_j \partial_k S^k_{\text{core}}(x)$. Applying the chain rule to $E(\sigma(x_{-k}))$,

$$\partial_j\big[E(\sigma(x_{-k}))\big] = E\big(\sigma(x_{-k})\big) \cdot \operatorname{sech}^2\left(\frac{\sigma(x_{-k})}{s_{\max}}\right) \cdot \partial_j \sigma(x_{-k}). \tag{55}$$

Substituting $\partial_j \sigma(x_{-k}) = \psi_j'(x_j)$ from (50) and factoring out $E(\sigma(x_{-k}))$ yields (53). Because ψ_j and ρ_j are unconstrained in sign, they provide unrestricted multiplicative and additive conditioning while strictly preserving $\partial_k S^k(x) \geq \epsilon_{\text{SM}} > 0$, establishing Criterion 4. □

5 Practical Enforcement Mechanisms

Having established in Section 4 that the deep monotonic KAN (Definition 4.7) and the Scale-Modulated Monotonic KAN (Definition 4.9) satisfy all four admissibility criteria of Definition 3.1, we now detail how these properties are enforced numerically:

1. **Target Monotonicity and Conditioning Freedom (Criterion 1 and Criterion 4):** Recalling Lemma 4.2, monotonicity of each 1D B-spline on its internal knot domain is governed entirely by discrete control-point increments $c_{m+1} - c_m$, which we enforce via an *unconstrained cumulative softplus reparameterization* (Section 5.1).

2. **Global Real-Line Domain (Criterion 2):** To guarantee that these derivative bounds hold across all of $\mathbb{R}^d$ whenever input samples or intermediate hidden activations fall outside the finite knot grid $[t_{\min}, t_{\max}]$, we attach C^1*-continuous linear tail extrapolations* to every univariate spline (Section 5.2).

3. **Higher Regularity and Analytic Derivative Computability (Criterion 3 and Key Requirement 3):** We parameterize the univariate edges using B-splines of degree $p \geq 3$, which guarantees the third-order differentiability required by the generalized precision integrand $T_{jk}(x)$ and enables exact, closed-form evaluation of all required derivatives via B-spline recurrence (Section 5.3).

5.1 Cumulative Softplus Reparameterization of B-Spline Control Points

By Lemma 4.2, a degree-p B-spline $\phi(t) = \sum_{m=1}^{M} c_m B_{m,p}(t)$ is non-decreasing on its active knot domain $[t_{p+1}, t_{M+1}]$ whenever its control coefficients satisfy $c_{m+1} - c_m \geq 0$, and strictly increasing with uniform derivative lower bound $\phi'(t) \geq \epsilon > 0$ whenever $c_{m+1} - c_m \geq \delta > 0$ for all $m \in \{1, \ldots, M-1\}$. Reparameterizing the control vector $c = (c_1, \ldots, c_M)^\top \in \mathbb{R}^M$ of each monotonic B-spline in terms of an unconstrained base coefficient $c_1 \in \mathbb{R}$ and unconstrained increment parameters $\theta = (\theta_1, \ldots, \theta_{M-1})^\top \in \mathbb{R}^{M-1}$,

$$c_{m+1} = c_m + \text{softplus}(\theta_m) + \delta_{\text{mono}}, \quad m = 1, \ldots, M-1, \tag{56}$$

or equivalently in closed cumulative-sum form,

$$c_m = c_1 + \sum_{r=1}^{m-1} \Big(\mathrm{softplus}(\theta_r) + \delta_{\mathrm{mono}} \Big), \quad m = 2, \dots, M, \tag{57}$$

where $\mathrm{softplus}(u) := \log(1 + e^u) > 0$ and $\delta_{\mathrm{mono}} \geq 0$ is a fixed increment floor.

Because $\mathrm{softplus}(u) > 0$ for all $u \in \mathbb{R}$ with $\inf_{u \in \mathbb{R}} \mathrm{softplus}(u) = 0$, every control-point increment satisfies,

$$c_{m+1} - c_m = \mathrm{softplus}(\theta_m) + \delta_{\mathrm{mono}} > \delta_{\mathrm{mono}} \quad \forall \theta_m \in \mathbb{R}. \tag{58}$$

Along the active backbone path $\mathcal{P}^* \in \mathbb{P}_k$ of Definition 4.7 (or the active path q^* in Definition 4.4), setting a strictly positive floor $\delta_{\mathrm{mono}} = \delta > 0$ (e.g., $\delta_{\mathrm{mono}} = 10^{-4}$) enforces condition (26) of Lemma 4.2 identically for all $\theta \in \mathbb{R}^{M-1}$, certifying $\phi'_{l,i_l^*,i_{l-1}^*}(t) \geq \epsilon_l > 0$. On the remaining target-transmitting splines where only non-decreasing slopes $\phi'_{l,j,i}(t) \geq 0$ are required, one may set either $\delta_{\mathrm{mono}} = 0$ or $\delta_{\mathrm{mono}} > 0$. In contrast, for all conditioning inner splines $\phi_{1,j,i}$ $(i \neq k)$ in Definition 4.7 and the scale and offset splines ψ_j, ρ_j $(j \neq k)$ in Definition 4.9, no reparameterization is applied: their control vectors $c \in \mathbb{R}^M$ are updated directly as free parameters, preserving unrestricted sign and curvature for conditioning (Criterion 4). Because the map $(c_1, \theta) \mapsto c$ in (57) is smooth (C^∞) and bijective onto $\{c \in \mathbb{R}^M : c_{m+1} - c_m > \delta_{\mathrm{mono}}\}$, standard unconstrained optimizers such as Adam or L-BFGS can be applied directly without projection steps.

5.2 Global C^1 Linear Tail Extrapolation

To satisfy Criterion 2, the monotonicity guarantees of Section 4 must hold on all of $\mathbb{R}^d$, even when a spline input $t \in \mathbb{R}$—whether a raw sample coordinate x_i at layer $l = 1$ or an intermediate hidden activation $x_i^{(l-1)} = \sum_{r=1}^{n_{l-2}} \phi_{l-1,i,r}(x_r^{(l-2)})$ at layer $l \geq 2$—falls outside the active knot interval $[t_{\min}, t_{\max}] := [t_{p+1}, t_{M+1}]$.

To guarantee well-posedness across all of $\mathbb{R}$, we extend every univariate spline ϕ outside $[t_{\min}, t_{\max}]$ via C^1-continuous linear tail extrapolation,

$$\phi_{\mathrm{extrap}}(t) = \begin{cases} \phi(t_{\min}) + \phi'(t_{\min})(t - t_{\min}), & t < t_{\min}, \\ \phi(t), & t_{\min} \leq t \leq t_{\max}, \\ \phi(t_{\max}) + \phi'(t_{\max})(t - t_{\max}), & t > t_{\max}, \end{cases} \tag{59}$$

whose first derivative is globally continuous and constant on both exterior tails,

$$\phi'_{\mathrm{extrap}}(t) = \begin{cases} \phi'(t_{\min}), & t < t_{\min}, \\ \phi'(t), & t_{\min} \leq t \leq t_{\max}, \\ \phi'(t_{\max}), & t > t_{\max}. \end{cases} \tag{60}$$

By Lemma 4.2 and the softplus reparameterization (56), the boundary slopes satisfy $\phi'(t_{\min}) \geq \epsilon_l > 0$ and $\phi'(t_{\max}) \geq \epsilon_l > 0$ for every spline along the active path $\mathcal{P}^*$, and $\phi'(t_{\min}) \geq 0$, $\phi'(t_{\max}) \geq 0$ for all remaining target-transmitting splines. Consequently, the derivative never vanishes in the tails, and the global positivity bounds of Theorems 4.5 and 4.8 and Proposition 4.10 hold uniformly across all of $\mathbb{R}^d$.

5.3 Higher-Order Regularity and Closed-Form Derivative Recurrence

To satisfy Criterion 3, the transport map $S^k(x)$ must possess well-defined derivatives up to third order so that the generalized precision integrand $T_{jk}(x)$ in (19) which depends on the mixed third

derivative $\partial_j \partial_k^2 S^k(x)$ can be evaluated smoothly. A univariate B-spline $\phi(t) = \sum_{m=1}^{M} c_m B_{m,p}(t)$ of polynomial degree p defined over simple interior knots is C^∞ on every open knot span (t_m, t_{m+1}) and belongs to continuity class C^{p-1} across the interior knots. Enforcing Criterion 3 therefore reduces to selecting a sufficiently high spline degree $p \geq 3$: cubic B-splines ($p = 3$) are C^2-continuous across interior knots with bounded, piecewise-constant third derivatives $\phi'''(t)$ (avoiding Dirac delta singularities at the knots), while degree $p \geq 4$ yields globally continuous third derivatives across the active knot domain.

Crucially, this B-spline parameterization simultaneously fulfills Key Requirement 3 (efficient derivative computability). Iterating the first-derivative recurrence (27) of Lemma 4.2 shows that the r-th derivative $\phi^{(r)}(t)$ $(1 \leq r \leq p)$ of a degree-p B-spline on $[t_{\min}, t_{\max}]$ is itself an explicit B-spline of degree $p - r$,

$$\phi^{(r)}(t) = \sum_{m=r+1}^{M} c_m^{(r)} B_{m,p-r}(t), \quad \text{where} \quad c_m^{(r)} = \frac{p-r+1}{t_{m+p-r+1} - t_m} \Big(c_m^{(r-1)} - c_{m-1}^{(r-1)} \Big), \tag{61}$$

initialized at $c_m^{(0)} = c_m$ (with $\phi_{\text{extrap}}^{(r)}(t) \equiv 0$ for all $r \geq 2$ on the linear tails $t \notin [t_{\min}, t_{\max}]$). Because each basis function $B_{m,p-r}(t)$ has compact support over $p - r + 1$ knot intervals, evaluating $\phi(t)$ together with $\phi'(t)$, $\phi''(t)$, and $\phi'''(t)$ requires only $\mathcal{O}(p)$ arithmetic operations in closed form. In stark contrast to UMNNs which must differentiate through a numerical quadrature rule via nested automatic differentiation. Our B-spline KAN parameterization yields exact analytic expressions for all partial derivatives of $S^k(x)$, which we now exploit to compute the generalized precision matrix in closed form.

6 Exact Analytic Derivatives for the Generalized Precision Matrix

Evaluating the empirical generalized precision matrix $\widehat{\Omega}_{jk} = \frac{1}{N} \sum_{i=1}^{N} |T_{jk}(x^{(i)})|$ from (17) requires the mixed partial derivatives of the conditional log-likelihood up to third order. We adopt the following shorthand notation:

$$S_k \coloneqq \partial_k S^k(x), \qquad S_j \coloneqq \partial_j S^k(x), \tag{62}$$

$$S_{kk} \coloneqq \partial_k^2 S^k(x), \qquad S_{jk} \coloneqq \partial_j \partial_k S^k(x), \tag{63}$$

$$S \coloneqq S^k(x), \qquad S_{jkk} \coloneqq \partial_j \partial_k^2 S^k(x), \tag{64}$$

for all $j \neq k$. Differentiating $T_{jk}(x) = \partial_j \partial_k [-\frac{1}{2} S^2 + \log S_k]$ via the product and quotient rules yields the decomposition,

$$T_{jk}(x) = \underbrace{-S_j S_k - S S_{jk}}_{\text{Score Coupling Term}} + \underbrace{\frac{S_{jkk} S_k - S_{kk} S_{jk}}{S_k^2}}_{\text{Hessian Curvature Term}}. \tag{65}$$

6.1 Closed-Form Core KAN Derivatives

For the two-layer monotonic KAN core $S_{\text{core}}^k(x) = \sum_{q=1}^{Q} \phi_{2,1,q}(x_q^{(1)}(x))$ with $x_q^{(1)}(x) = \sum_{i=1}^{d} \phi_{1,q,i}(x_i)$ (Definition 4.4), repeated application of the chain and product rules expresses all five partial derivatives required by (65) directly in terms of the univariate B-spline derivatives $\phi^{(r)}$ $(r \in \{1, 2, 3\})$

from (61),

$$S_{k,\mathrm{core}}(x) = \sum_{q=1}^{Q} \phi'_{2,1,q}\big(x_q^{(1)}\big)\, \phi'_{1,q,k}(x_k), \tag{66}$$

$$S_{j,\mathrm{core}}(x) = \sum_{q=1}^{Q} \phi'_{2,1,q}\big(x_q^{(1)}\big)\, \phi'_{1,q,j}(x_j), \tag{67}$$

$$S_{jk,\mathrm{core}}(x) = \sum_{q=1}^{Q} \phi''_{2,1,q}\big(x_q^{(1)}\big)\, \phi'_{1,q,j}(x_j)\, \phi'_{1,q,k}(x_k), \tag{68}$$

$$S_{kk,\mathrm{core}}(x) = \sum_{q=1}^{Q} \Big[\phi''_{2,1,q}\big(x_q^{(1)}\big)\big(\phi'_{1,q,k}(x_k)\big)^2 + \phi'_{2,1,q}\big(x_q^{(1)}\big)\, \phi''_{1,q,k}(x_k)\Big], \tag{69}$$

$$S_{jkk,\mathrm{core}}(x) = \sum_{q=1}^{Q} \phi'_{1,q,j}(x_j)\Big[\phi'''_{2,1,q}\big(x_q^{(1)}\big)\big(\phi'_{1,q,k}(x_k)\big)^2 + \phi''_{2,1,q}\big(x_q^{(1)}\big)\, \phi''_{1,q,k}(x_k)\Big]. \tag{70}$$

All five quantities are evaluated in closed form via vectorized tensor contractions across $j \in \{1,\dots,d\}\backslash\{k\}$ in a single forward pass, without numerical quadrature or automatic differentiation graphs.

6.2 Scale-Modulated Derivatives and the Hessian Cancellation Theorem

For the Scale-Modulated Monotonic KAN $S^k(x) = E \cdot S^k_{\mathrm{core}}(x) + \alpha(x_{-k})$ (Definition 4.9), where $E \coloneqq E(\sigma(x_{-k})) = \exp(s_{\max}\tanh(\sigma(x_{-k})/s_{\max}))$, $\sigma_j \coloneqq \partial_j \log E = \operatorname{sech}^2(\sigma(x_{-k})/s_{\max})\psi'_j(x_j)$, and $\partial_j\alpha(x_{-k}) = \rho'_j(x_j)$ for $j \neq k$, differentiating with respect to x_k and x_j yields,

$$S_k \coloneqq E\, S_{k,\mathrm{core}}, \qquad S_j \coloneqq E\big(\sigma_j S_{\mathrm{core}} + S_{j,\mathrm{core}}\big) + \rho'_j, \tag{71}$$

$$S_{kk} \coloneqq E\, S_{kk,\mathrm{core}}, \qquad S_{jk} \coloneqq E\big(\sigma_j S_{k,\mathrm{core}} + S_{jk,\mathrm{core}}\big), \tag{72}$$

$$S \coloneqq E\, S_{\mathrm{core}} + \alpha, \qquad S_{jkk} \coloneqq E\big(\sigma_j S_{kk,\mathrm{core}} + S_{jkk,\mathrm{core}}\big). \tag{73}$$

Theorem 6.1 (Hessian Cancellation Under Scale Modulation)**.** *Under the Scale-Modulated Monotonic KAN parameterization of Definition 4.9, the Hessian curvature term of the generalized precision integrand (65) is invariant to both the conditioning scale $\sigma(x_{-k})$ and the conditioning offset $\alpha(x_{-k})$,*

$$\frac{S_{jkk}S_k - S_{kk}S_{jk}}{S_k^2} = \frac{S_{jkk,core}S_{k,core} - S_{kk,core}S_{jk,core}}{(S_{k,core})^2}, \tag{74}$$

and the full precision integrand reduces to,

$$T_{jk}(x) = -S_jS_k - SS_{jk} + \frac{S_{jkk,core}S_{k,core} - S_{kk,core}S_{jk,core}}{(S_{k,core})^2}. \tag{75}$$

Proof. Substituting S_k, S_{kk}, S_{jk}, and S_{jkk} from (71)–(73) into the Hessian quotient gives,

$$\begin{aligned}\frac{S_{jkk}S_k - S_{kk}S_{jk}}{S_k^2} &= \frac{E\big(\sigma_j S_{kk,\mathrm{core}} + S_{jkk,\mathrm{core}}\big)\cdot ES_{k,\mathrm{core}} - ES_{kk,\mathrm{core}}\cdot E\big(\sigma_j S_{k,\mathrm{core}} + S_{jk,\mathrm{core}}\big)}{E^2(S_{k,\mathrm{core}})^2} \\ &= \frac{E^2\Big(\sigma_j S_{kk,\mathrm{core}}S_{k,\mathrm{core}} + S_{jkk,\mathrm{core}}S_{k,\mathrm{core}} - \sigma_j S_{kk,\mathrm{core}}S_{k,\mathrm{core}} - S_{kk,\mathrm{core}}S_{jk,\mathrm{core}}\Big)}{E^2(S_{k,\mathrm{core}})^2}.\end{aligned} \tag{76}$$

The scale-gradient cross-terms $\pm\sigma_j S_{kk,\mathrm{core}}S_{k,\mathrm{core}}$ cancel identically in the numerator, and canceling the common positive factor $E^2 > 0$ yields (74). Substituting (74) into (65) gives (75). □

6.3 Vectorized Precision Evaluation via Tensor Contractions

In our PyTorch implementation (`models/kan_transport.py`), all core derivatives (66)–(70) and scale-modulated quantities (71)–(73) are evaluated simultaneously across a batch of B samples $x \in \mathbb{R}^{B\times d}$ using batched tensor contractions (`torch.einsum`):

1. **Spline Basis Evaluation:** For each variable coordinate $j \in \{1,\dots,d\}$, evaluate B-spline basis matrices:
$$b_{\text{val}},\, b_{d1},\, b_{d2},\, b_{d3} \in \mathbb{R}^{B\times d\times M}. \tag{77}$$

2. **Inner Layer Sums and Edge Derivatives:**
$$u_q = \texttt{torch.einsum("bdm,dqm->bq", b_val, inner_coeffs)}, \tag{78}$$
$$\phi'_{b,j,q} = \texttt{torch.einsum("bdm,dqm->bdq", b_d1, inner_coeffs)}, \tag{79}$$
$$\phi''_{b,j,q} = \texttt{torch.einsum("bdm,dqm->bdq", b_d2, inner_coeffs)}. \tag{80}$$

3. **Outer Spline Evaluations on Hidden Activations u_q:**
$$\Phi'_q = \texttt{torch.einsum("bqm,qm->bq", Phi_d1, outer_coeffs)}, \tag{81}$$
$$\Phi''_q = \texttt{torch.einsum("bqm,qm->bq", Phi_d2, outer_coeffs)}, \tag{82}$$
$$\Phi'''_q = \texttt{torch.einsum("bqm,qm->bq", Phi_d3, outer_coeffs)}. \tag{83}$$

4. **Vectorized Core Partial Derivatives:**
$$S_{k,\text{core}} = \texttt{torch.einsum("bq,bq->b", Phi_p, phi_k_p)}, \tag{84}$$
$$S_{j,\text{core}} = \texttt{torch.einsum("bq,bdq->bd", Phi_p, phi_prime)}, \tag{85}$$
$$S_{jk,\text{core}} = \texttt{torch.einsum("bq,bdq,bq->bd", Phi_pp, phi_prime, phi_k_p)}, \tag{86}$$
$$S_{kk,\text{core}} = \texttt{torch.einsum("bq,bq->b", Phi_pp, phi_k_p**2)} + \texttt{torch.einsum("bq,bq->b", Phi_p, phi_k_pp)}, \tag{87}$$
$$S_{jkk,\text{core}} = \texttt{torch.einsum("bdq,bq->bd", phi_prime, term_inner)}. \tag{88}$$

Because every derivative is computed in closed form across all conditioning dimensions $j \in \{1,\dots,d\}\setminus\{k\}$ in a single vectorized pass, extracting the generalized precision matrix $\widehat{\Omega}$ completely bypasses autodiff graphs, achieving a 110.8× speedup over `torch.func.hessian` and an 84× to 300× speedup over UMNN numerical quadrature.

7 Summary, Empirical Benchmarks, and Conclusions

The computational efficiency, memory scaling, and structural expressivity of the proposed Scale-Modulated Monotonic KAN (SM-KAN) compared to the baseline UMNN architecture are summarized in Table 1.

Table 1: Head-to-head comparison between baseline UMNN and Scale-Modulated KAN (SM-KAN).

Metric / Property	**UMNN (Baseline)**	**Scale-Modulated KAN (Ours)**
Numerical Quadrature	Required (50 nodes/sample)	**None** (Exact closed-form)
Precision Evaluation ($d=10$)	10.56 s	**0.14 s** (84× faster)
Precision Evaluation ($d=40$)	203.6 s	**0.68 s** (299× faster)
Precision Evaluation ($d=156$, Ovarian)	≈ 42 min (projected)	**1.99 s** (> 1200× faster)
Peak Memory Footprint ($d=10$)	71.1 MB	**0.25 MB** (284× reduction)
Target Monotonicity ($\partial_k S^k > 0$)	Certified ($f > 0$)	**Certified** (Proposition 4.10)
Parameters per Feature ($d=10$)	1,523	**485** (3.14× more compact)
Training Throughput	33,011 samples/s	**104,591 samples/s** (3.17× faster)
Multiplicative Dependencies	Supported (slow autodiff)	**Fully Supported** ($F_1 = 1.0$)

7.1 Benchmark Performance on Standard L-SING Testbeds

We evaluate SM-KAN against the baseline UMNN architecture across the three primary benchmark domains introduced in the original L-SING paper [6]:

1. **Gaussian Graphical Model ($d = 10$):** Evaluated on $M = 5{,}000$ training and $N = 2{,}000$ evaluation samples from a 10-dimensional Gaussian distribution with 4 ground-truth edges. At conservative threshold $\tau = 0.20$, both UMNN and SM-KAN achieve perfect recovery ($F_1 = 1.000, \mathrm{FPR} = 0.000$). However, at threshold $\tau = 0.10$, UMNN degrades to $F_1 = 0.667$ with 4 false positives (FPR $= 0.098$), whereas SM-KAN maintains **perfect graph recovery ($F_1 = 1.000, \mathbf{FPR} = 0.000$)**. Training SM-KAN requires 28.1 s compared to 89.9 s for UMNN (3.2× faster), and precision matrix extraction completes in 0.14 s versus 10.56 s (84× faster).

2. **Non-Gaussian Butterfly Distribution ($d = 10$ and $d = 40$):** To evaluate heteroscedastic scale dependencies ($X_i \sim \mathcal{N}(0, 1)$, $Y_i = W_i X_i$), we test recovery on $d = 10$ (5 independent pairs) across sample sizes $M \in \{1{,}000, 5{,}000\}$. At $M = 5{,}000$, SM-KAN achieves $F_1 = 1.000$ and FPR $= 0.000$ across all thresholds $\tau \in [0.10, 0.30]$, maintaining a large separation margin ($\Delta = 0.303$) between the weakest true edge (0.415) and the strongest spurious non-edge (0.112). At small sample size $M = 1{,}000$, UMNN incurs 18 false positives ($F_1 = 0.357, \mathrm{FPR} = 0.450$), whereas SM-KAN maintains substantially higher precision ($F_1 = 0.625, \mathrm{FPR} = 0.150$). Scaling to $d = 40$ (20 pairs, 780 undirected candidate edges), extracting the complete precision matrix takes **0.68 seconds** for SM-KAN compared to **203.6 seconds** for UMNN (299× speedup).

3. **TCGA Curated Ovarian Cancer Gene Network ($d = 156$):** Demonstrating applicability to high-dimensional biological networks, we train SM-KAN across 156 gene expression profiles from patient tumor samples ($N_{\mathrm{trn}} = 346, N_{\mathrm{val}} = 115, N_{\mathrm{tst}} = 117$). All 156 conditional transport maps train in 254.9 s total, and evaluating the entire 156×156 generalized precision matrix across test patients takes only **1.99 seconds** with a memory footprint of 75.4 MB.

7.2 Summary of Contributions

In summary, the Scale-Modulated Monotonic KAN parameterization resolves the fundamental computational and architectural limitations of existing transport maps in L-SING:

1. **Elimination of Numerical Quadrature:** By replacing ODE/integral neural networks with B-splines, SM-KAN computes forward evaluations, functional losses, and precision matrices in closed form, delivering 84× to 299× faster precision extraction.

2. **Unconditional Monotonicity by Construction:** Monotonicity $\partial_k S^k(x) \geq \epsilon > 0$ is certified globally on $\mathbb{R}^d$ through cumulative softplus reparameterization of discrete control points and C^1 linear tail extrapolation, eliminating post-step projections.

3. **Resolution of the Additive Polarization Dilemma:** B-spline scale modulation $\exp(\sigma(x_{-k})) \cdot S^k_{\mathrm{core}}(x) + \alpha(x_{-k})$ enables the map to represent complex multiplicative and heteroscedastic dependencies while unconditionally preserving target monotonicity.

A Location–Scale Decoupling and Mixed Couplings in SM-KAN

In this appendix, we detail why the Scale-Modulated Monotonic KAN (Definition 4.9),

$$S^k(x) = \exp\big(\sigma(x_{-k})\big) \cdot S^k_{\mathrm{core}}(x) + \alpha(x_{-k}), \tag{89}$$

includes an explicit post-multiplication conditioning offset $\alpha(x_{-k}) = \sum_{j \neq k} \rho_j(x_j)$ in addition to the conditioning splines already present inside $S^k_{\text{core}}(x)$.

A.1 Heteroscedastic Location–Scale Butterfly Example

In the standard Butterfly distribution (44), $X_2 = W \cdot X_1$ with $W \sim \mathcal{N}(0,1)$ and $X_1 \perp W$, the conditional distribution $X_2 \mid (X_1 = x_1) \sim \mathcal{N}(0, x_1^2)$ is centered at zero ($\mathbb{E}[X_2 \mid X_1 = x_1] = 0$). Only its standard deviation $|x_1|$ varies with x_1, so normalizing X_2 to $\mathcal{N}(0,1)$ requires pure rescaling by $1/|x_1|$ with $\alpha(x_1) = 0$.

More generally, suppose X_2 exhibits both a conditional mean shift $\mu(X_1)$ and a multiplicative conditional scale,

$$X_2 = \mu(X_1) + W \cdot X_1, \quad X_1 \sim \mathcal{N}(0,1), \quad W \sim \mathcal{N}(0,1), \quad X_1 \perp W. \tag{90}$$

Conditioned on $X_1 = x_1 \neq 0$, we have $X_2 \mid (X_1 = x_1) \sim \mathcal{N}(\mu(x_1), x_1^2)$, and the exact Knothe–Rosenblatt transport map normalizing X_2 to $\mathcal{N}(0,1)$ is an affine function of x_2,

$$S^2(x_1, x_2) = \frac{x_2 - \mu(x_1)}{|x_1|} = \underbrace{\exp\big(-\log|x_1|\big)}_{\exp(\sigma(x_1))} \cdot \underbrace{x_2}_{S^2_{\text{core}}(x_1,x_2)} + \underbrace{\left(-\frac{\mu(x_1)}{|x_1|}\right)}_{\alpha(x_1)}. \tag{91}$$

Here each component of SM-KAN fulfills a distinct role: $S^2_{\text{core}}(x_1, x_2) = x_2$ provides the strictly increasing dependence on the target coordinate x_2 ($\partial_2 S^2_{\text{core}} = 1 > 0$), $\exp(\sigma(x_1)) = 1/|x_1| > 0$ rescales the target slope to normalize the conditional variance, and $\alpha(x_1) = -\mu(x_1)/|x_1|$ shifts the intercept so that the transformed reference variable $Z_2 = S^2(X_1, X_2)$ has zero mean.

A.2 Why $\alpha(x_{-k})$ Is Placed Outside the Product

Although $S^k_{\text{core}}(x)$ already accepts x_{-k} through its inner conditioning splines $\phi_{1,q,j}(x_j)$, placing the additive offset $\alpha(x_{-k}) = \sum_{j \neq k} \rho_j(x_j)$ outside the multiplication by $\exp(\sigma(x_{-k}))$ decouples additive location shifts from multiplicative scaling across multiple conditioning variables ($d > 2$). Every term inside $S^k_{\text{core}}(x)$ is multiplied by the full scale factor $\exp(\sigma(x_{-k})) = \prod_{j \neq k} \exp(\psi_j(x_j))$. If coordinate x_1 modulates the conditional variance of X_3 while coordinate x_2 only shifts the conditional mean of X_3, placing the shift from x_2 inside S^3_{core} forces its contribution in z_3-space to be multiplied by $\exp(\psi_1(x_1))$. Including $\alpha(x_{-k}) = \sum_{j \neq k} \rho_j(x_j)$ outside the product allows each conditioning coordinate x_j to contribute an unscaled additive location shift $\rho_j(x_j)$ directly to $S^k(x)$, independent of $\exp(\sigma(x_{-k}))$.

Furthermore, separating $\alpha(x_{-k})$ from the product prevents large inverse scales from amplifying centering errors. In regions where the conditional variance is small (such as near $x_1 \approx 0$ in the Butterfly distribution), the inverse scale factor $\exp(\sigma(x_1)) \approx 1/|x_1|$ becomes large. If the conditional mean shift had to be represented entirely inside $S^k_{\text{core}}(x)$, any small approximation error in the offset $S^k_{\text{core}}(x_1, 0)$ would be magnified by $\exp(\sigma(x_1)) \gg 1$. Placing $\alpha(x_{-k})$ after the multiplication adjusts the center directly in the standardized $\mathcal{N}(0,1)$ reference space without amplification by $\exp(\sigma(x_{-k}))$.

A.3 Unscaled Target Reduction and Simultaneous Additive–Multiplicative Couplings

A natural question regarding the SM-KAN parameterization (49) is whether multiplying $S^k_{\text{core}}(x)$ by $\exp(\sigma(x_{-k}))$ forces the target coordinate x_k to be multiplicatively coupled to every conditioning coordinate x_j ($j \neq k$), even when the true dependency is purely additive or combines an additive parent with a multiplicative parent.

Two algebraic properties ensure that x_k is never spuriously scaled by purely additive conditioning coordinates. First, because $\sigma(x_{-k}) = \sum_{j\neq k} \psi_j(x_j)$ enters through the exponential map, the scale factor decomposes as a coordinate-wise product,

$$\exp\big(\sigma(x_{-k})\big) = \prod_{j\neq k} \exp\big(\psi_j(x_j)\big). \tag{92}$$

For any conditioning coordinate x_j that does not modulate the conditional variance of X_k, setting $\psi_j(x_j) \equiv 0$ yields the multiplicative identity $\exp(\psi_j(x_j)) = \exp(0) \equiv 1$. In particular, when all relations on node k are purely additive, $\sigma(x_{-k}) \equiv 0$ gives $\exp(\sigma(x_{-k})) \equiv 1$, reducing (49) identically to the unscaled additive map $S^k(x) = S^k_{\text{core}}(x) + \alpha(x_{-k})$.

Second, suppose a target variable X_6 depends simultaneously on an additive parent X_5 and a multiplicative parent X_3,

$$X_6 = cX_5 + X_3 \cdot Z_6, \quad Z_6 \sim \mathcal{N}(0,1), \quad Z_6 \perp (X_3, X_5). \tag{93}$$

Inverting (93) to recover the independent Gaussian reference variable $Z_6 = S^6(x)$ yields,

$$S^6(x) = \frac{x_6 - cx_5}{|x_3|} = \underbrace{\exp\big(-\log|x_3|\big)}_{\exp(\psi_3(x_3))} \cdot \big(\underbrace{x_6 - cx_5}_{S^6_{\text{core}}(x)}\big), \quad x_3 \neq 0. \tag{94}$$

With $\psi_5(x_5) \equiv 0$ ($\exp(\psi_5(x_5)) \equiv 1$), the additive parent x_5 never multiplies x_6; instead, subtracting the conditional mean cx_5 and dividing the entire residual $(x_6 - cx_5)$ by $|x_3|$ reproduces the exact transport map (94).

A.4 Six-Dimensional Mixed Coupling Benchmark

To verify this decoupling behavior empirically, we construct a 6-dimensional synthetic benchmark graph comprising three distinct coupling regimes simultaneously,

$$\text{Purely Additive Pair } (1,2): \quad X_1 = Z_1, \quad X_2 = 1.2\sin(X_1) + Z_2, \tag{95}$$

$$\text{Purely Multiplicative Pair } (3,4): \quad X_3 = Z_3, \quad X_4 = X_3 \cdot Z_4, \tag{96}$$

$$\text{Mixed Additive + Multiplicative } (3,5,6): \quad X_5 = Z_5, \quad X_6 = 1.2X_5 + X_3 \cdot Z_6, \tag{97}$$

where $Z_1, \ldots, Z_6 \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)$. When evaluated on this benchmark, SM-KAN successfully recovers all four true graph edges ($(1,2)$, $(3,4)$, $(5,6)$, and $(3,6)$) while keeping spurious cross-couplings suppressed below threshold (< 0.19).

Inspecting the empirical activations of the learned scale splines ψ_j and offset splines ρ_j confirms that SM-KAN dynamically decouples location shifts from scale modulation without supervision:

1. On the additive node X_2, the scale spline ψ_1 remains flat ($\text{std}(\psi_1) = 0.031$, so $\exp(\psi_1) \approx 1$), and the nonlinear mean shift $\sin(X_1)$ is routed entirely through the offset spline ρ_1 ($\text{std}(\rho_1) = 0.137$).

2. On the multiplicative node X_4, the offset spline is inactive ($\text{std}(\rho_3) = 0.014$), and the heteroscedastic scale is captured by the scale spline ($\text{std}(\psi_3) = 0.165$).

3. On the mixed node X_6, SM-KAN selectively assigns the multiplicative parent X_3 to the scale spline ($\text{std}(\psi_3) = 0.181$ vs. $\text{std}(\rho_3) = 0.063$) and the additive parent X_5 to the offset spline ($\text{std}(\rho_5) = 0.161$ vs. $\text{std}(\psi_5) = 0.024$). This achieves superior test NLL (-0.136 for SM-KAN vs. $+0.107$ for purely additive KAN) and confirms that purely additive coordinates are never spuriously scaled.

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